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Professional Summary

Software Engineer with expertise in signal processing, machine learning, image processing, and system optimization. Skilled in Java, C++, and Python with a proven ability to develop and deploy high-performance software solutions. Passionate about advancing high precision equipment and AI in industrial applications.

Education

University of Michigan Ann - Arbor *August 2021 - April 2023*
Master of Science in Computer Engineering – Signal & Image Processing and Machine Learning GPA: 3.72/4.00

University of Illinois at Urbana - Champaign *August 2017 - May 2021*
Bachelor of Science in Computer Engineering (Honors) GPA: 3.71/4.00

Professional Experience

Software Engineer

KLA Corporation - Milpitas, CA *January 2023 - Present*

- Design, develop, and maintain software that powers cutting-edge wafer inspection machines.
- Develop code for machine movement and fully automated calibration sequences.
- Integrate complex hardware and firmware systems to ensure seamless interaction.
- Address and resolve field escalations with customers, occasionally within Class 100 cleanroom environments.
- Leverage multi-threading techniques in **Java**, **Python**, and **C++** to deliver scalable solutions, improving throughput for next-generation machines.

Software Engineering Intern

Siemens Digital Industries Software - Ann Arbor, MI *January 2022 - August 2022*

- Prototyped new releases and automated testing processes for NX CAD freeform modeling tools.
- Implemented parametrization intersections, surface boundary intersections and sub-sampling with 2-D and 3-D mapping.
- Enhanced system capabilities using **C++**, incorporating mesh modeling, reverse engineering, and geometric computation.

Research and Development Intern

Shure Inc. - Niles, IL *June 2020 - August 2020*

- Deployed pre-trained models on dedicated IoT embedded systems for Machine Learning inferences.
- Analyzed model graphs using Netron and performed graph surgery for conversion and quantization for deployment on the IoT boards.

Key Projects

Hackathon - Intelligent Digital Emulator Application

KLA Corporation *August 2024*

- Developed a machine learning model to generate synthetic image patterns for simulating focus metrics and calibrations.
- Utilized transfer learning techniques to enhance model adaptability across various optical states of the machine.
- Reduced on-tool developer time for calibration related features by 75% by emulating image collection using this model and integrating into simulator builds.

Reinforcement Learning Independent Study with Prof. Lei Ying

University of Michigan *August 2022 - April 2023*

- Researched self-supervised learning techniques to enhance data efficiency in Reinforcement Learning.
- Extended Data Efficient SSL research for RL by analyzing the efficacy of temporal effects in contrastive and SPR loss.

Intrinsic Reward with Contrastive Random Walk (*arXiv:2204.10976*)

University of Michigan *February 2022 – April 2022*

- Applied CRW-based curiosity models to accelerate policy optimization in reinforcement learning environments.
- Demonstrated faster convergence and higher reward acquisition compared to traditional curiosity methods.

Leadership and Activities

Corporate Relations Officer

Eta Kappa Nu (HKN) - Beta Epsilon *January 2022 - April 2023*

- Organized networking events, connecting students with industry professionals through tech talks and workshops.
- Facilitated collaboration between corporate sponsors and the university.

Skills and Interests

Spoken Languages: Native proficiency in English and Hindi

Programming Languages: Java, Python, C/C++, Matlab

Tools & Systems: Linux, Git, IBM Engineering Workflow Management, MySQL

Hobbies: Rock Climbing, Swimming, Guitar, Badminton, Hiking, Photography, Sketching, Origami